



# Electrical Machines

Detailed Solutions

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**GATE INSTRUMENTATION ENGINEERING**

**PREPFUSION**

Previous Year Questions Solutions

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## SOLUTIONS



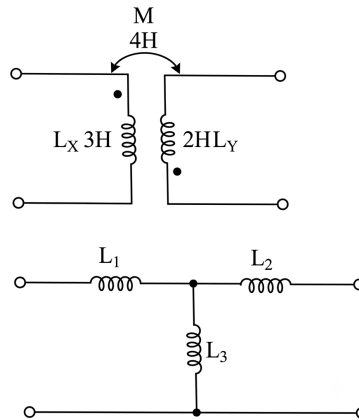
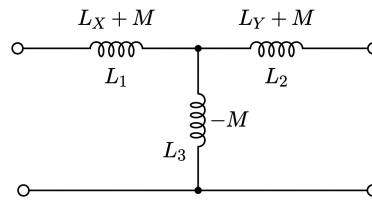
# Magnetic Circuits

**Topics**

1. Magnetically Coupled Circuits, Dot Convention
2. Energy in Coupled circuits
3. Electric and Magnetic Materials

**Q1.25.1**
**MCQ**
**2025**
**1M**
**Ans: A**


Given circuit is in a magnetically opposing dot configuration. By modeling the magnetically coupled coils into its equivalent T-network:


**Fig. Solution Q1.25.1**


Equivalent circuit

**Fig. Solution Q1.25.1**

For a mutually coupled inductor pairs with opposing dots, the mutual terms add up positively in the series arms of the equivalent T-network ( $L_1 = L_X + M$ ,  $L_2 = L_Y + M$ ) and negatively in the common shunt branch ( $L_3 = -M$ ). The values of the equivalent T-model inductances can be derived directly by analyzing the network:

$$L_1 = L_X + M = 3 + 4 = 7 \text{ H}$$

$$L_2 = L_Y + M = 2 + 4 = 6 \text{ H}$$

$$A) L_1 = 7 \text{ H}, \quad L_2 = 6 \text{ H}, \quad L_3 = -4 \text{ H}$$

**Q1.24.1**
**NAT**
**2024**
**2M**
**Ans: 0.25**


Let the two self-inductances be equal such that:

$$L_1 = L_2 = L$$

When connected in a series-aiding configuration, the equivalent total inductance is given by:

$$L_1 + L_2 + 2M = 500 \text{ mH} \implies 2L + 2M = 500 \text{ mH} \quad \text{--- (1)}$$

When connected in a series-opposing configuration, the equivalent total inductance is given by:

$$L_1 + L_2 - 2M = 300 \text{ mH} \implies 2L - 2M = 300 \text{ mH} \quad \text{--- (2)}$$

Adding equations (1) and (2):

$$2(L_1 + L_2) = 800 \text{ mH} \implies 2(2L) = 800 \text{ mH}$$

$$L = L_1 = L_2 = 200 \text{ mH}$$

Substituting  $L = 200 \text{ mH}$  back into equation (1):

$$2(200) + 2M = 500 \implies 2M = 100 \implies M = 50 \text{ mH}$$

The coupling coefficient  $K$  is calculated as:

$$K = \frac{M}{\sqrt{L_1 L_2}} = \frac{M}{L} = \frac{50 \text{ mH}}{200 \text{ mH}} = \frac{1}{4} = 0.25$$

$$\boxed{0.25}$$

**Q1.23.1**

**NAT**

**2023**

**2M**

**Ans: 17.37**



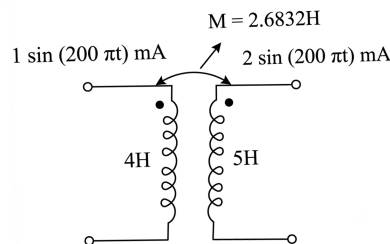
Given,

$$K = 0.6$$

$$L_1 = 4 \text{ H}, \quad L_2 = 5 \text{ H}$$

$$i_1(t) = 1 \sin(200\pi t) \text{ mA} \implies I_{1P} = 1 \text{ mA}$$

$$i_2(t) = 2 \sin(200\pi t) \text{ mA} \implies I_{2P} = 2 \text{ mA}$$



**Fig. Solution Q1.23.1**

Mutual inductance  $M$ :

$$M = K\sqrt{L_1 L_2} = 0.6\sqrt{4 \times 5} = 0.6\sqrt{20} \approx 2.6832 \text{ H}$$

The maximum peak energy stored in the mutually coupled circuit occurs when both currents reach their peak values simultaneously (since they are in phase). Looking at the dot polarity, both currents enter the dots, meaning the magnetic fields aid each other:

$$E_{\text{Peak}} = \frac{1}{2}L_1 I_{1P}^2 + \frac{1}{2}L_2 I_{2P}^2 + M I_{1P} I_{2P}$$

Substituting peak values in milliamperes ( $1 \text{ mA} = 10^{-3} \text{ A}$ ):

$$E_{\text{Peak}} = \frac{1}{2}(4)(1 \times 10^{-3})^2 + \frac{1}{2}(5)(2 \times 10^{-3})^2 + 2.6832(1 \times 10^{-3})(2 \times 10^{-3})$$

$$E_{\text{Peak}} = [2 \times 10^{-6}] + [10 \times 10^{-6}] + [5.3664 \times 10^{-6}] \text{ J}$$

$$E_{\text{Peak}} = (2 + 10 + 5.3664) \mu\text{J} = 17.3664 \mu\text{J}$$

Rounding off to two decimal places:

$$\boxed{17.37}$$

### Key Points

- Total instantaneous energy stored in coupled coils:  $E(t) = \frac{1}{2}L_1i_1^2 + \frac{1}{2}L_2i_2^2 \pm Mi_1i_2$
- The sign of the  $M$  term is positive if both currents enter (or both exit) the dot terminals.

### Mistakes to be avoided

- Do not forget to convert milliamperes to amperes ( $\times 10^{-3}$ ), which becomes  $10^{-6}$  (micro) when squared or multiplied.

**Q1.15.1**

MCQ

2015

1M

Ans: B



From the ideal transformer voltage ratio relations:

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

$$\frac{V_2}{V_1} = \frac{1}{N}$$

$$V_2 = NV_1 \implies NV_i \sin \omega t$$

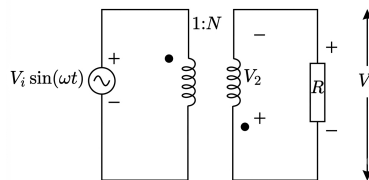


Fig. Solution Q1.15.1

According to the given dot polarity convention, the current enters the dot on the primary winding while the induced voltage is positive at the terminal opposite to the dot on the secondary winding. Thus, the polarity of  $V_2$  is inverted relative to the marked terminal orientation of  $V_o$ :

$$V_o = -NV_i \sin \omega t$$

$$B) - NV_i \sin \omega t$$

**Q1.13.1**

MCQ

2013

2M

Ans: B



We evaluate the circuit arrangement under two distinct excitation setups:

**Case 1:**  $V_{wx1} = 100 \text{ V}$  is applied across WX terminals. The secondary side voltage of the ideal transformer with a turns ratio of 1 : 1.25 becomes:

$$V_{\text{secondary}} = 100 \times 1.25 = 125 \text{ V}$$

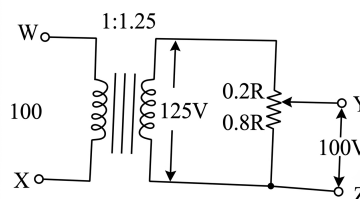


Fig. Solution Q1.13.1

The attenuator network splits the resistance into a  $0.2R$  and  $0.8R$  potentiometer configuration. The open-circuit voltage across the output terminals YZ is taken across the  $0.8R$  section:

$$V_{yz1} = 125 \times \frac{0.8R}{0.2R + 0.8R} = 125 \times 0.8 = 100 \text{ V}$$

Hence, the first ratio is:

$$\frac{V_{yz1}}{V_{wx1}} = \frac{100}{100}$$

**Case 2:**  $V_{yz2} = 100 \text{ V}$  is applied backward across YZ terminals. Since it is an ideal transformer, it draws zero magnetizing current to set up the rated flux ( $i = 0$ ). Consequently, there is no current flow and no voltage drop across the upper attenuator resistor segment  $0.2R$ :

$$V_{\text{secondary}} = V_{yz2} = 100 \text{ V}$$

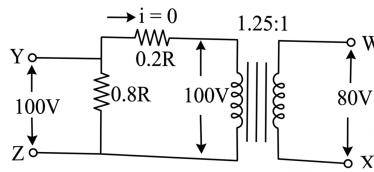


Fig. Solution Q1.13.1

Stepping down the voltage to the primary winding using the effective 1.25 : 1 reverse turns relation:

$$V_{wx2} = \frac{100}{1.25} = 80 \text{ V}$$

Hence, the second ratio is:

$$\frac{V_{wx2}}{V_{yz2}} = \frac{80}{100}$$

$$\boxed{B) 100/100 \text{ and } 80/100}$$

### Q1.13.1

MCQ

2013

2M

Ans: C



The individual quality factors ( $Q$ -factors) for the two uncoupled inductive coils at a chosen operating angular frequency  $\omega$  for a series RL circuit are defined as:

$$q_1 = \frac{\omega L_1}{R_1}, \quad q_2 = \frac{\omega L_2}{R_2}$$

When these two coils are connected in series, the total equivalent inductance  $L_t$  and total equivalent resistance  $R_t$  become:

$$L_t = L_1 + L_2$$

$$R_t = R_1 + R_2$$

The effective quality factor  $q_t$  of the series combination at the same operating frequency is given by:

$$q_t = \frac{\omega(L_1 + L_2)}{R_1 + R_2}$$

Substituting  $\omega L_1 = q_1 R_1$  and  $\omega L_2 = q_2 R_2$  into the equation:

$$q_t = \frac{q_1 R_1 + q_2 R_2}{R_1 + R_2}$$

$$\boxed{C) (q_1 R_1 + q_2 R_2) / (R_1 + R_2)}$$

# DEBUG INFO

## Chapter I

Magnetic Circuits

**Total Questions : 6**

MCQ : 4

MSQ : 0

NAT : 2



## SOLUTIONS



# Single Phase Transformers

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**Topics**

1. Equivalent Circuit and Phasor Diagram
2. Open Circuit and Short Circuit Tests
3. Voltage Regulation and Efficiency of Transformer

**Q2.26.1**
**MCQ**
**2026**
**1M**
**Ans: C**
☒ ☐ ☐

The secondary impedance referred to the primary side is:

$$Z'_2 = \frac{Z_2}{K^2}, \quad \left( K = \frac{N_2}{N_1} \right)$$

$$Z'_2 = \left( \frac{N_1}{N_2} \right)^2 \cdot Z_2$$

For a step-down transformer:

$$\frac{N_1}{N_2} > 1$$

$$\therefore Z'_2 > Z_2$$

$$\boxed{C) Z'_2 > Z_2}$$

**Q2.26.2**
**NAT**
**2026**
**2M**
**Ans: 2490 to 2510**
☒ ☒ ☐

Given parameters:

$$\cos \phi = \frac{P}{VI} = \frac{363}{220 \times 2.75} = 0.6$$

The magnetizing current component is:

$$I_m = I_0 \sin \phi = 2.75 \times 0.8 = 2.2 \text{ A}$$

The magnetizing reactance  $X_m$  relative to the low-voltage side is:

$$X_m = \frac{V}{I_m} = \frac{220}{2.2} = 100 \Omega$$

Referred to the high-voltage side, the magnetizing reactance becomes:

$$X'_m = \frac{X_m}{K^2} = \frac{100}{\left( \frac{220}{1100} \right)^2}$$

$$= \left( \frac{1100}{220} \right)^2 \times 100 = 2500 \Omega$$

$$\boxed{2500}$$

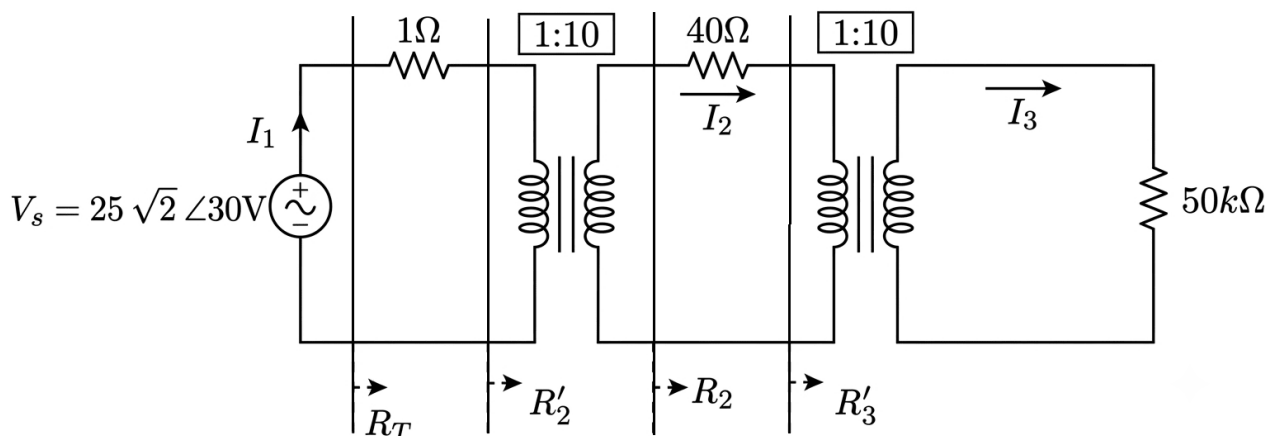
**Q2.25.1**
**NAT**
**2025**
**2M**
**Ans: 31.00 to 31.50**
☒ ☒ ☐


Fig. Solution Q2.25.1

The transformer network can be systematically referred across stages by squaring the respective turns ratios. Referring the loads back step-by-step simplifies the multi-stage network into a single equivalent loop:

$$R_{eq} = 1 \Omega + 4 \Omega = 5 \Omega$$

The loop current is found using the RMS voltage of the source:

$$I = \frac{25}{10} (\text{rms}) \implies I = 2.5 \text{ A}$$

The power dissipated in the load is:

$$P_{50 \text{ k}\Omega} = P_{5 \Omega} = 2.5^2 \times 5 = 31.25 \text{ W}$$

$$\boxed{31.25}$$

**Q2.24.1** **NAT** **2024** **2M** **Ans: 95.2 to 95.4** **● ● ○**

Given parameters: Rating = 50 kVA,  $\eta_{FL, UPF} = 95\%$ , pf = 1 and Core losses = neglected ( $W_i = 0$ )

Using the full-load efficiency equation at unity power factor:

$$W_{cu}(x) = x^2 W_{cu, FL}$$

$$\text{Efficiency} = \frac{\text{Output Power}}{\text{Output Power} + \text{Iron Loss} + \text{Copper Loss}}$$

$$\% \eta_{FL, UPF} = \frac{50 \times 10^3 \times 1 \times 1}{(50 \times 10^3 \times 1 \times 1) + (W_i = 0) + W_{cu, FL}}$$

$$0.95 = \frac{50 \times 10^3}{(50 \times 10^3) + W_{cu, FL}} \implies W_{cu, FL} = 2631.57 \text{ W}$$

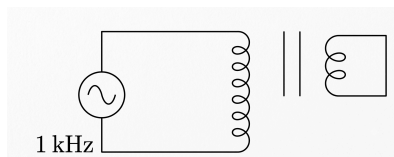
Now computing efficiency at 75% full load and 0.8 power factor lagging:

$$\begin{aligned} \% \eta_{75\%FL, 0.8pf} &= \frac{(0.75 \times 50 \times 10^3) \times 0.8}{(0.75 \times 50 \times 10^3 \times 0.8) + 0 + (0.75)^2 \times 2631.57} \times 100 \\ &= 95.29\% \end{aligned}$$

$$\boxed{95.2 \text{ to } 95.4}$$

**Q2.23.1** **NAT** **2023** **2M** **Ans: 1** **● ● ●**

Given short-circuit test data at  $f = 1 \text{ kHz}$ :  $w_{sc} = 8 \text{ W}$ ,  $V_{sc} = 6 \text{ V}$ ,  $I_{sc} = 2 \text{ A}$



**Fig. Solution Q2.23.1**

The short-circuit impedance and equivalent parameters referred to the primary are:

$$Z_{sc} = \frac{V_{sc}}{I_{sc}} = \frac{6}{2} = 3 \Omega$$

$$w_{sc} = I_{sc}^2 R_{01} \implies R_{01} = \frac{8}{2^2} = 2 \Omega$$

$$Z_{sc}^2 = R_{01}^2 + X_{01}^2 \implies X_{01}^2 = 3^2 - 2^2 = 5$$

$$X_{01} = \sqrt{5} \implies X_{01} = +j\sqrt{5}$$

When connected with a capacitor  $C$  across a 2 V source:

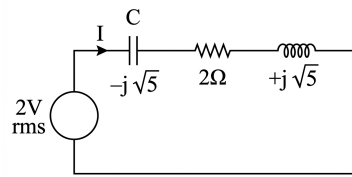


Fig. Solution Q2.23.1

The capacitance value is:

$$C = \frac{1}{2\pi\sqrt{5}} \text{ mF}$$

The capacitive reactance is:

$$-jX_c = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 10^3 \times \frac{1}{2\pi\sqrt{5}} \times 10^{-3}} = -jX_c = \sqrt{5}$$

Evaluating the final circuit loop at resonance: Since  $X_L = X_C$ , the circuit is purely resistive and under a resonance condition:

$$I = \frac{V}{R} = \frac{2}{2} = 1 \text{ A}$$

1

Q2.21.1

NAT

2021

1M

Ans: 0.24 to 0.26

● ● ○

Given parameters:  $L_0 = 250 \times 10^{-3} \text{ H}$ ,  $R_0 = 300 \Omega$  The inductive reactance is:

$$X_{L_0} = 2\pi f L_0 = 2\pi \times 50 \times 250 \times 10^{-3} = 78.54 \Omega$$

The susceptance and conductance are:

$$B_{L_0} = \frac{1}{X_{L_0}} = \frac{1}{78.54} = 0.01273 \text{ U}$$

$$G_0 = \frac{1}{300} = 3.33 \times 10^{-3} \text{ U}$$

The total admittance in parallel circuit is:

$$Y_0 = \sqrt{G_0^2 + B_{L_0}^2} = 0.01316 \text{ U}$$

The no-load power factor is:

$$\cos \phi_0 = \frac{G_0}{Y_0} = 0.253 \text{ (lag)}$$

0.253

**Q2.21.2****MCQ****2021****2M****Ans: A**

Given parameters:  $\eta_{\max} = 98\%$ ,  $W_{\text{iron}} = 80 \text{ W}$ ,  $R_{01} = 0.5 \Omega$ ,  $I_1 = 25 \text{ A}$  At maximum efficiency, the variable copper loss equals the constant iron loss ( $x$  is the loading fraction):

$$x^2 I_1^2 R_{01} = W_i$$

$$x^2 (25)^2 (0.5) = 80$$

$$x = 50.59\%$$

Thus, maximum efficiency occurs at 50.59% of full load.

A) 50.6%

# DEBUG INFO

## Chapter II

Single Phase Transformers

**Total Questions : 7**

MCQ : 2

MSQ : 0

NAT : 5

## SOLUTIONS



# Three Phase Induction Machines

**Topics**

1. Principle of Operation , Types and Performance
2. Equivalent Circuit of Induction Motor
3. Torque-Slip, Torque-Speed Characteristics
4. Stability and Testing – No-Load (OC) and Blocked-Rotor (SC) Tests
5. Starting of Induction Motors
6. Speed Control
7. Types of losses and efficiency calculation of Electric Machines

**Q3.25.1**
**MCQ**
**2025**
**2M**
**Ans: A**


Given conditions:

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{2} = 3000 \text{ rpm}$$

The initial slip is:

$$s_1 = \frac{3000 - 2970}{3000} = 0.01$$

In the normal operating (low-slip) region of a standard induction motor, torque varies linearly with slip.

$$T_1 \propto s_1$$

As torque is doubled,  $T_2 = 2T_1$ , meaning the slip gets doubled as well:

$$S_2 = 2S_1 = 2 \times 0.01 = 0.02$$

Evaluating the new rotor speed  $N_2$ :

$$S_2 = \frac{N_s - N_2}{N_s} \implies \frac{3000 - N_2}{3000} = 0.02$$

$$3000 - N_2 = 60 \implies N_2 = 2940 \text{ rpm}$$

A) 2940

**Q3.24.1**
**NAT**
**2024**
**2M**
**Ans: 17.00 to 17.30**


Given:

$$T_{st} = 100\% \text{ of } T_{FL} \implies \frac{T_{st}}{T_{FL}} = 1$$

$$T_{max} = 300\% \text{ of } T_{FL} \implies \frac{T_{max}}{T_{FL}} = 3$$

$$S_{T_{max}} = ?$$

The ratio of starting torque to maximum torque is a specific function of the parameter  $a$ , where  $a$  corresponds to the slip at which maximum torque occurs ( $s_m$ ).

Using the torque ratios:

$$\therefore \frac{T_{st}/T_{FL}}{T_{max}/T_{FL}} = \frac{1}{3} \implies \frac{T_{st}}{T_{max}} = \frac{1}{3} = \frac{2a}{1+a^2}$$

$$\frac{2a}{1+a^2} = \frac{1}{3} \implies a^2 + 1 - 6a = 0$$

$$a^2 - 6a + 1 = 0$$

Solving for  $a$  (slip at maximum torque):

$$a = 0.1715 = 17.15\%$$

17.15

**Q3.23.1**
**NAT**
**2023**
**1M**
**Ans: 0**


Output voltage when machine is rotated in the clockwise direction is 200 V. If it is rotated in the opposite direction, the armature current direction and shunt field current direction will be changed. Then  $\phi_{sh}$  opposes the  $\phi_{residual}$ . Net EMF will become zero.

0



**Q3.22.1****NAT****2022****1M****Ans: 1440**☒ ☐ ☐

Given parameters:  $V = 440 \text{ V}$ ,  $P = 8 \text{ kW}$ , Poles = 4,  $f = 50 \text{ Hz}$ ,  $S = 0.04$  The synchronous speed is calculated as:

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

The actual rotor speed  $N_r$  is:

$$N_r = N_s(1 - s) = 1500(1 - 0.04) = 1440 \text{ rpm}$$

1440

**Q3.21.1****MCQ****2021****2M****Ans: A**☒ ☒ ☐

By adding external resistance in the rotor circuit of a slip ring induction motor, the total rotor resistance increases, which subsequently shifts the maximum torque point and increases the starting torque. This improves the starting performance of a wound-rotor induction motor by bringing the starting torque closer to the maximum available breakdown torque.

A) The starting torque would be higher.

## DEBUG INFO

### Chapter III

Three Phase Induction Machines

**Total Questions : 5**

MCQ : 2

MSQ : 0

NAT : 3